Euclidean Geometry In Mathematical Olympiads

EGMO Chapter 1 Problem 1.47 (IMO 2006/1)

i just realized how much easier incenter/excenter was -.- but at least this didn't take too long either

or i could've just stopped when I found AIP was obtuse

This post has been edited 3 times. Last edited by mathlogician, Mar 24, 2020, 9:25 AM

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ike.chen
1162 posts

Jun 14, 2021, 12:11 AM #10

Terrible Old Solution

The given angle condition implies

$$\angle PBC + \angle PCB = \frac{1}{2}(\angle B + \angle C)$$

so

$$\angle BPC = 90^\circ + \frac{\angle A}{2}.$$

Hence, we know B, I, P, C are concyclic.

By the Incenter-Excenter Lemma, we know the center of $(BIPC)$ is the midpoint of arc BC, which we denote by M. Since A, I, M_a are collinear, we conclude the circle centered at A with radius AI is tangent to $(BIPC)$. Since I is in between A and M, the result follows easily. $\blacksquare$

Remark: We don't need to confirm that quadrilateral $BIPC$ is convex. (This is a common misconception.)

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serpent_121
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Jun 21, 2021, 10:36 PM • 1 #11

Let ray CI intersect segment AB at F, and let ray BI intersect segment AC at E. Then I think it might make sense to note at the start of the solution that point P must within one of $\triangle BIF$ or $\triangle CIE$, but not on any of the segments FI or IE. This ensures that $BPIC$ is actually a quadrilateral and eliminates any configuration issues; I also think gives you more intuition about the $P = I$ case.

Chesssaga
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Jun 25, 2021, 11:25 AM #12

Synthetic sol

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